

Amaan Khan

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Research Interests

Machine learning for robotics and autonomous systems, with a focus on safe and reliable deployment of learned models, spanning perception, planning, and control.

Education

Washington University in St. Louis

Master of Science in Computer Engineering Expected 2027
Graduate Certificate in Machine Learning and Data Mining

Washington University in St. Louis

Bachelor of Science in Computer Science 2025

Research Experience

Haptic Interface Systems for Motor Vehicle Piloting May 2026 - Present

Digital Transformation Research Corps, WashU

Advisors: Dr. Shantanu Chakrabartty and Jonathan Hanahan

Developing a haptic-guided driving simulator to explore non-visual data-delivery strategies and reduce cognitive overload in compromised environments.

Trustworthy Autonomous Systems Lab Jan 2026 - Present

Washington University in St. Louis, CSE Dept.

Advisor: Dr. Hussein Sibai

Constructing class- and distance-conditioned conformal contracts for agent state estimation in semantic occupancy perception, and embedding these into robust safety constraints to enable lightweight, calibration-based adaptation to deployment-domain shifts without retraining the underlying perception stack.

Cake-Cutting Fairness Explorations May 2025 - Aug 2025

Washington University in St. Louis, REU Program

Advisor: Dr. Ron Cytron

Analyzed computational complexity and game-theoretic properties of fair-division algorithms and developed pedagogical implementations for Mozilla's Responsible Computer Science initiative.

Teaching Experience

Washington University in St. Louis

Department of Computer Science and Engineering

Teaching Assistant Positions:

CSE 468T: Introduction to Quantum Computing (Two semesters) 2025 - Present

CSE 131: Introduction to Computer Science (Three semesters) 2022–2023

Department of Mechanical Engineering and Materials Science

MEMS 202: Introduction Computer-Aided Design (One semester) 2022

Activities & Service

WashU AI Racing 2026 - Present

Perception Team, ML Engineer

WashU Graduate and Professional Council 2026 - Present

Computer Science Department Representative

WashU Robotics 2024–Present

SWARM Team, Software Engineer

Awards & Honors

Research Grant, Adaption Labs inaugural cohort (2026) — selected to investigate test-time adaptation of Vision-Language-Action models for novel robotic manipulation tasks without weight updates.

CMD-IT/ACM Richard Tapia Celebration of Diversity in Computing Conference Scholarship, 2025. Full scholarship from Washington University McKelvey School of Engineering